

Math 472: Homework 05

Due Friday, February 27

Problem 1 (Exercise 8.31). In a study to compare the perceived effects of two pain relievers, 200 randomly selected adults were given the first pain reliever, and 93% indicated appreciable pain relief. Of the 450 individuals given the other pain reliever, 96% indicated experiencing appreciable relief.

- (a) Give an estimate for the difference in the proportions of all adults who would indicate perceived pain relief after taking the two pain relievers. Provide a bound on the error of estimation.
- (b) Based on your answer to part (a), is there evidence that proportions experiencing relief differ for those who take the two pain relievers? Why?

Problem 2 (Exercise 8.17). If Y has a binomial distribution with parameter n and p , the $\hat{p} = \frac{Y}{n}$ is an unbiased estimator of p . Another estimator is $\hat{q} = \frac{Y+1}{n+2}$.

- (a) What is the bias of $\hat{q}$?
- (b) Derive $\text{MSE}(\hat{p})$ and $\text{MSE}(\hat{q})$.
- (c) For what values of p is $\text{MSE}(\hat{p}) < \text{MSE}(\hat{q})$?

Problem 3 (Exercise 8.39). Suppose the random variable Y has a gamma distribution with shape parameter $\alpha = 2$ and unknown scale parameter β (see section 4.6 in textbook). It can be shown that $2Y/\beta$ has a chi-squared distribution with 4 degrees of freedom (you don't have to show this).

- (a) Using $2Y/\beta$ as a pivotal quantity, derive a 90% confidence interval of β .
- (b) Using R, test your confidence interval by generating 10,000 samples $Y_1, \dots, Y_{10000}$ (each with a randomly chosen numerical value for β) and counting the proportion of times that your confidence interval contains the true value of β . (It should be close to 90%).

Problem 4 (Exercise 8.40). Suppose that the random variable Y is an observation from a normal distribution with unknown mean μ and variance 1.

- (a) Find a 95% confidence interval for μ .
- (b) Find a 95% upper confidence interval for μ .
- (c) Find a 95% lower confidence interval for μ .

Problem 5 (Exercise 8.41). Suppose that Y is normally distributed with mean $\mu = 0$ and unknown variance σ^2 . Then Y^2/σ^2 has a chi-squared distribution with 1 degree of freedom. Use the pivotal quantity Y^2/σ^2 to find

- (a) A 95% confidence interval for σ^2 .
- (b) A 95% upper confidence interval for σ^2 .
- (c) A 95% lower confidence interval for σ^2 .

Problem 6 (Part of exercise 6.18). The Pareto distribution is a “power-law” distribution used for modeling things like income distributions and earthquake magnitudes. Specifically, given a random variable X , we say that distribution of X is a member of the *Pareto family of distributions* if its distribution function is

$$F_X(x) = \mathbb{P}[X \leq x] = \begin{cases} 1 - \left(\frac{\beta}{x}\right)^\alpha & : x \geq \beta \\ 0 & : x < \beta \end{cases} \quad (1)$$

for some positive parameters α and β .

- (a) Big-O notation: Let f and g be real-valued functions whose domains include the positive real numbers. We write $f(x) = O(g(x))$ as $x \rightarrow \infty$ if there exist constants $M, k > 0$ such that

$$\frac{|f(x)|}{|g(x)|} \leq M$$

for all $x > k$. (This is read as “ f is big-O of g ”.)

Show that $\mathbb{P}[X \geq x] = O(\frac{1}{x^\alpha})$ as $x \rightarrow \infty$. (In words, this says that the upper tail of the probability distribution of X decays like $\frac{1}{x^\alpha}$.)

- (b) Derive the density function of X . Make sure to specify the function on all parts of its domain: both when $x \geq \beta$ and when $x < \beta$.
- (c) Show that the improper integral $\int_1^\infty \frac{1}{x^\delta} dx$ diverges to $+\infty$ if $\delta \leq 1$, but converges to $\frac{1}{\delta-1}$ if $\delta > 1$.
- (d) What is $\mathbb{E}[X]$? Consider the cases $\alpha \leq 1$ and $\alpha > 1$ separately.

Problem 7 (Part of exercise 9.28). Let $X_1, \dots, X_n$ be a random sample from the distribution F_X (defined in Eq. (1)), and let T be the statistic $T(X_1, \dots, X_n) = \min(X_1, \dots, X_n)$.

- (a) Show that the sampling distribution of T is

$$F_T(t) = \begin{cases} 1 - \left(\frac{\beta}{t}\right)^{\alpha n} & : t \geq \beta \\ 0 & : t < \beta \end{cases}. \quad (2)$$

(Hint: For every real number t , we have $[\min(X_1, X_2, \dots, X_n) > t] = [X_1 > t, X_2 > t, \dots, X_n > t]$.)

- (b) Derive the density function of T from Eq. (2). Make sure to specify the function on all parts of its domain: both when $t \geq \beta$ and when $t < \beta$.
- (c) Qualitatively describe what happens to the graph of f_T as n becomes very large. Sketch this situation, and interpret your picture: is T likely to be close to β or far from β when n is large?

Problem 8 (Exercise 8.15). Let $X_1, \dots, X_n$ be a random sample from the Pareto distribution with probability density function

$$f_X(x) = \begin{cases} 3\beta^3 x^{-4} & : x \geq \beta \\ 0 & : x < \beta \end{cases},$$

where $\beta > 0$ is unknown. (So X is a Pareto distribution with $\alpha = 3$). We shall consider the estimator $\hat{\beta} = T(X_1, \dots, X_n) = \min(X_1, \dots, X_n)$.

- (a) What is the bias of $\hat{\beta}$? Simplify your answer.
- (b) What is $\text{MSE}(\hat{\beta})$? You don't need to simplify your answer.